

Adaptive Hard-Pair Sampling via Curriculum Learning for Speech Separation

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Abstract

End-to-end neural speech separation systems have achieved strong average scale-invariant signal-to-distortion ratio (SI-SDR) scores, yet still struggle with mixtures where speakers have highly similar timbre. These difficult cases reduce robustness and raise concerns about real-world generalization. To address this, we propose a curriculum hard-pair sampling strategy. During training, a pairwise speaker-speaker difficulty matrix is built and updated online based on the SI-SDR score of separating each speaker pair. Each mixture for model training is constructed by first randomly sampling an anchor speaker, and the interference speaker is sampled based on a temperature-controlled Softmax which gives more weights to speakers that are, as indicated by the difficulty matrix, difficult to be separated with the anchor speaker. Results on Libri2Mix show that the proposed method consistently improves performance, especially for similar-timbre mixtures, without additional training overhead.

Index Terms: speech separation, dynamic mixing, hard-pair sampling

1. Introduction

Monaural speech separation aims to recover the speech signal of each speaker from a single-channel multi-speaker mixture [1–15]. It is a core problem to solve in automatic speech recognition, meeting transcription, and voice interaction [16–19]. Despite rapid progress in neural networks based models, performance still degrades for mixtures with heavy overlap, similar voice timbre, low signal-to-noise ratio (SNR), and strong reverberation. Popular models such as Conv-TasNet [20], DPRNN [21] and TF-GridNet [22] are trained by uniformly sampling training mixtures and typically optimizing the mean SI-SDR metric [23]. As a result, rare but difficult mixtures inevitably receive less emphasis during training, leaving large errors on a small subset of mixtures. These errors undermine the robustness of the models in real applications, and often dominate downstream failures and users' perception of the performance of the separation models.

When the speakers in a mixture have very similar timbre, current models tend to produce large residual interference [24, 25]. We refer to these similar speaker pairs as *hard pairs* and all other pairs as *easy pairs*. Hard pairs typically form only a small portion of the training data, yet accounting for most tail errors. They lower overall scores and, more importantly, harm generalization to real recordings. However, common training

configurations sample mixtures uniformly, so the network sees much fewer hard pairs than easy pairs, and therefore is prone to overfit the many easy pairs. As a result, the network would underperform when similar voices occur. This motivates training strategies that place more focus on rare but important hard pairs.

To deal with this issue, Wang *et al.* [26] and Tzinis *et al.* [27] emphasize hard mixtures by re-weighting losses or gradients. While this can reduce errors on hard examples, it inevitably alters the effective training distribution and often sacrifices mean performance on the original test distribution. For the task of target speaker extraction, Liu *et al.* [28] propose a curriculum that gradually increases target–interferer similarity, but the schedule is hand-crafted rather than driven by the model's actual errors, making its hyper-parameters difficult to tune. In addition, both the global hard-mining strategy proposed by Wang *et al.* [26] and the curriculum proposed by Liu *et al.* [28] require pre-computing difficulty scores and partitioning the entire training set into easy and hard subsets, which becomes cumbersome and time-consuming on large corpora. Differently, the curriculum hard-pair sampling algorithm proposed in this paper maintains an online speaker–speaker difficulty matrix that is updated during training based on the SI-SDR scores of each speaker pair. For each training mixture, we sample an anchor speaker uniformly and then select the interferer via a temperature-controlled Softmax that gradually shifts from uniform to hard-biased sampling. This design automatically separates easy and hard pairs on the fly, avoids loss-level distribution shift, preserves average performance, and introduces essentially no additional computational overhead.

Compared with uniform sampling, our proposed curriculum, motivated by [29], focuses learning on rare but important cases without letting them dominate from the start. Compared with the approach by Wang *et al.* [26], which increases the presence of hard mixtures through fixed re-weighting or re-sampling, our schedule adapts the hardness over time, and balances easy and hard cases more naturally. In practice, this yields better lower tail performance while preserving mean SI-SDR, since the model first builds a strong base on easy pairs and then learns to resolve mixtures with similar timbres as the proportion of hard pairs rises.

2. Proposed Method

We consider monaural speech separation. Let $x \in \mathbb{R}^N$ denote the observed mixture signal and $\{s_c\}_{c=1}^C$ the underlying C clean speaker signals. The mixture is modeled as $x = \sum_{c=1}^C s_c$. Let $\mathcal{S}$ denote the set of speakers in the training set. Each training mixture is generated from a speaker tuple $(i_1, \dots, i_C)$ with $i_k \in \mathcal{S}$, and the separation model f_θ maps the mixture x to estimated sources $f_\theta(x) = \{\hat{s}_c\}_{c=1}^C$. The model parameters θ

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are optimized to maximize a separation metric such as SI-SDR based on all training mixtures.

In many speech separation studies [22, 30–35], speaker tuples are sampled uniformly from a pre-mixed set of mixtures provided with the tasks or from mixtures created via dynamic mixing on the fly, and the model is trained on this fixed sampling distribution. This uniform sampling strategy treats all the mixtures equally, regardless of their difficulty (for example, heavy overlap or similar timbre). As a result, during training, rare but difficult combinations would occupy only a small fraction of updates, and large errors can persist on a small subset of hard mixtures.

In this work, we replace (static) uniform sampling with a dynamic curriculum over speaker pairs. During training, the sampler adapts the probability of selecting each speaker combination based on an online difficulty estimate (derived from SI-SDR scores or related statistics) so that early epochs stay close to uniform sampling for stable, initial learning, while later epochs gradually allocate more emphasis on hard pairs. This section describes the construction of the difficulty measure and the curriculum hard-pair sampling scheme.

2.1. Online Speaker-Speaker Difficulty Matrix Construction

Let $\mathcal{S} = \{s_1, \dots, s_N\}$ denote the set of speakers for model training. We maintain a dense *speaker–speaker difficulty matrix* $\mathbf{D} \in \mathbb{R}^{N \times N}$, where each entry $D_{i,j}$ stores a running estimate of the separation difficulty for mixtures involving speaker pair (s_i, s_j) . The diagonal entries are ignored, since a speaker is never paired with itself. In addition, we keep a counting matrix $\mathbf{C} \in \mathbb{N}^{N \times N}$ that records how many times each pair has been observed during training. At the beginning of training, we initialize

$$D_{i,j} = d_0, \quad C_{i,j} = 0, \quad \forall i \neq j, \quad (1)$$

where d_0 is an undefined or neutral value (e.g., NaN or a constant), and we set $D_{i,i}$ for each $i \in \{1, \dots, N\}$ to be invalid. A fixed mapping between speaker IDs and matrix indices (i, j) is constructed once from the training corpus and re-used throughout the training process.

Whenever a mini-batch contains a mixture generated from speaker pair (s_i, s_j) , we obtain a scalar difficulty score $h_{i,j}$ for this mixture, such as the (negative) SI-SDR of the estimated sources. We then update the corresponding matrix entry by exponential moving average (EMA)

$$D_{i,j} \leftarrow \begin{cases} h_{i,j}, & C_{i,j} = 0, \\ (1 - \alpha) D_{i,j} + \alpha h_{i,j}, & C_{i,j} > 0, \end{cases} \quad (2)$$

where $\alpha \in (0, 1]$ is a smoothing factor, and we enforce symmetry $D_{i,j} = D_{j,i}$ and $C_{i,j} = C_{j,i}$. The count matrix is increased as

$$C_{i,j} \leftarrow C_{i,j} + 1, \quad C_{j,i} \leftarrow C_{j,i} + 1. \quad (3)$$

This online update requires only a constant amount of additional computation per observed speaker pair (a few scalar operations) and does not change the forward or backward complexity of the separation model. In contrast to offline hard-mining schemes that pre-compute difficulty scores for the entire corpus [26, 28], our matrix is updated on the fly and gradually reflects the model’s evolving notion of which speaker pairs are easy or hard to separate. In the next subsection, we show how the difficulty matrix $\mathbf{D}$ can be used to drive a curriculum over speaker pairs via a temperature-controlled sampling distribution.

2.2. Curriculum Hard-Pair Sampling

Given the online difficulty matrix $\mathbf{D}$, we describe how it is used to construct a curriculum over speaker pairs. For clarity, we focus on two-speaker cases (i.e., $C = 2$), where each training mixture is generated from an unordered pair of speakers (s_i, s_j) with $i \neq j$. Let $\mathcal{P} = \{(i, j) : 1 \leq i < j \leq N\}$ denote the set of all possible speaker pairs. Standard uniform sampling corresponds to drawing pairs (i, j) from $\mathcal{P}$ with equal probability, regardless of their difficulty. Instead of sampling pairs directly from $\mathcal{P}$, we decompose the process into two steps. First, we sample an anchor speaker i uniformly from $\{1, \dots, N\}$ so that every speaker has the same chance to serve as an anchor. Then, conditioned on the chosen anchor speaker i , we select a partner speaker (i.e., interference) $j \neq i$ according to a distribution that depends on the current difficulty estimates $\{D_{i,j}\}_{j=1, j \neq i}^N$. Let $\mathcal{C}_i \subseteq \{1, \dots, N\} \setminus i$ be the set of partners for which the difficulty entry $D_{i,j}$ has been observed at least once, that is, the corresponding count $C_{i,j}$ in the counting matrix is positive. For anchors with no previously observed partners (i.e., empty $\mathcal{C}_i$), we fall back to uniform sampling over $\{1, \dots, N\} \setminus i$.

For an anchor i with at least one observed partner, we treat $D_{i,j}$ as a difficulty score for pair (i, j) and define a temperature-controlled Softmax distribution over $\mathcal{C}_i$,

$$p_\tau(j|i) = \frac{\exp(D_{i,j}/\tau)}{\sum_{k \in \mathcal{C}_i} \exp(D_{i,k}/\tau)}, \quad \text{with } j \in \mathcal{C}_i, \quad (4)$$

where $\tau > 0$ is a temperature hyper-parameter. When τ is large, the distribution $p_\tau(j|i)$ approaches uniform over $\mathcal{C}_i$ and all partners are sampled with similar probability. As τ decreases, the distribution becomes increasingly concentrated on pairs with larger $D_{i,j}$, that is, pairs that are currently estimated to be hard for the model.

To provide a curriculum from easy to hard pairs, we let the temperature depend on the training epoch e . In our implementation, we use a simple linear schedule with a lower bound

$$\tau(e) = \max(\tau_{\min}, \tau_0(1 - \frac{e - 1}{E_{\text{decay}}))), \quad (5)$$

where τ_0 is the initial temperature, $\tau_{\min}$ is the minimum temperature, and E_{decay} controls the duration of the decay. At the beginning of training (i.e., small e), we have $\tau(e) \approx \tau_0$ and Eq. (4) is close to uniform, so the sampler behaves similarly to standard uniform pair sampling and the model can establish a strong baseline based on easy and moderately-difficult mixtures. As the training progresses, $\tau(e)$ gradually decreases towards $\tau_{\min}$ and the distribution (4) concentrates on harder pairs, exposing the model more frequently to mixtures that are difficult to separate, such as those with similar timbre.

We apply the curriculum training to a middle stage of training. In the early epochs, the model is still under-trained, so the SI-SDR scores do not yet provide a reliable estimate of pair difficulty. We therefore use only standard dynamic mixing at the beginning of training. In the final stage of training, we switch back to standard dynamic mixing, since continuously over-sampling the hardest pairs may over-bias the model toward a small hard tail and degrade its performance on easy pairs. Let E_{warm} and E_{stop} respectively denote the warm-up and stop epochs. For epochs $e \leq E_{\text{warm}}$ or $e > E_{\text{stop}}$, we simply draw pairs (i, j) from $\mathcal{P}$ uniformly, ignoring the difficulty matrix. For epochs $E_{\text{warm}} < e \leq E_{\text{stop}}$, we use the anchor–partner scheme with temperature $\tau(e)$ defined in Eq. (5). In our experiments, we set $E_{\text{warm}} = 20$, $E_{\text{stop}} = 100$.

The curriculum hard-pair sampling is implemented entirely at the data-sampling level. It incurs negligible additional cost. For each mini-batch, the extra computation consists of reading a row of the difficulty matrix for the chosen anchor, applying the Softmax operation in Eq. (4), and drawing a categorical sample. The forward and backward passes of the separation network remain unchanged. Together with the online update rule in Section 2.1, this design allows the sampler to automatically track the evolving notion of pairwise difficulty and to allocate more updates to hard mixtures, without modifying the loss function or introducing additional network components.

3. Experimental Setup

We evaluate the effectiveness of different models using the Libri2Mix dataset [36], a standard benchmark for two-speaker speech separation constructed based on clean speech signals in LibriSpeech [37] and noise signals in WHAM! [38]. In our experiments, we follow the Libri2Mix configuration and use both the 100 h and 360 h training subsets. According to the proprietary specification, the two subsets contain 13,900 and 50,800 two-speaker mixtures with 251 and 921 speakers, respectively, yielding a total of 64,700 mixtures and 1,172 distinct speakers. Instead of relying on pre-mixed audio, we perform on-the-fly dynamic mixing [30]. For each training example we randomly select two speakers from `train-clean-{100,360}`, pick one utterance per speaker, and mix them at 16 kHz. The inter-speaker SNR is drawn uniformly from the range $[-5, 5]$ dB.

To verify whether our proposed algorithm can be effective for different speech separation models, we employ TF-GridNet [22], Conv-TasNet [20] and BSRNN [39] as the backbone separation networks. All models are trained with the Adam optimizer [40], using an initial learning rate of 10^{-3} , which is halved if the validation performance does not improve for 4 consecutive epochs. For STFT, we use a square-root of the Hann window, with a window size of 16 ms and a hop size of 8 ms.

4. Evaluation Results

4.1. Effectiveness of Curriculum Hard-Pair Sampling

We evaluate the impact of curriculum hard-pair sampling on Libri2Mix. For each backbone model, we compare a dynamic-mixing baseline with uniform speaker-pair sampling against several curriculum variants with different initial sampling temperatures. All systems share the same training configuration, so that any performance difference can be attributed to the sampling strategy.

Table 1 reports the results for Conv-TasNet. The dynamic-mixing baseline without hard-pair sampling achieves an average SI-SDRi of 16.7 dB. When curriculum hard-pair sampling is enabled, low initial temperatures with $\tau_0 = 10$ and $\tau_0 = 15$ slightly degrade the mean SI-SDRi to 16.6 and 16.5 dB, respectively. With a higher initial temperature $\tau_0 = 20$, the proposed sampler matches the baseline at 16.7 dB. This indicates that overly-aggressive deviation from uniform sampling can hurt the average performance, whereas a milder curriculum that keeps the sampling distribution close to uniform in early epochs can preserve the mean SI-SDRi.

Table 2 shows the corresponding results for BSRNN. The dynamic-mixing baseline attains 21.0 dB SI-SDRi. Again, low initial temperatures ($\tau_0 = 10$ and $\tau_0 = 15$) lead to slightly worse averages of 20.6 and 20.9 dB. In contrast, using $\tau_0 = 20$ not only avoids degradation but improves the mean SI-SDRi

Table 1: Results of hard-pair sampling for Conv-TasNet on Libri2Mix.

DNN	Dynamic mixing?	Hard-pair sampling?	Init temp.	SI-SDRi (dB)↑
Conv-TasNet	✗	✗	-	14.8
Conv-TasNet	✓	✗	-	16.7
Conv-TasNet	✓	✓	10	16.6
Conv-TasNet	✓	✓	15	16.5
Conv-TasNet	✓	✓	20	16.7

Table 2: Results of hard-pair sampling for BSRNN on Libri2Mix.

DNN	Dynamic mixing?	Hard-pair sampling?	Init temp.	SI-SDRi (dB)↑
BSRNN	✗	✗	-	18.8
BSRNN	✓	✗	-	21.0
BSRNN	✓	✓	10	20.6
BSRNN	✓	✓	15	20.9
BSRNN	✓	✓	20	21.3

Table 3: Results of hard-pair sampling for TF-GridNet on Libri2Mix.

DNN	Dynamic mixing?	Hard-pair sampling?	Init temp.	SI-SDRi (dB)↑
TFGridNet	✓	✗	-	21.9
TFGridNet	✓	✓	10	22.3
TFGridNet	✓	✓	15	22.0
TFGridNet	✓	✓	20	22.7

to 21.3 dB, yielding a gain of 0.3 dB over the baseline. This suggests that the proposed curriculum is particularly beneficial when combined with a stronger backbone, and that a relatively high initial temperature strikes a balance between maintaining coverage of easy mixtures and increasingly focusing on hard speaker pairs.

Table 3 shows the corresponding results for TF-GridNet. Similar trend as in the BSRNN case is observed.

These results show that curriculum hard-pair sampling can be integrated into existing separation pipelines without sacrificing average performance, and can even improve the mean SI-SDRi when the initial temperature is chosen appropriately. In the following experiments, we fix the initial temperature to $\tau_0 = 20$ unless otherwise noted, and focus on analyzing the performance at various difficulty levels of input mixture.

4.2. SDRi Distribution and Worst-Case Performance

To understand how curriculum hard-pair sampling affects the performance at different difficulty levels, we analyze the distribution of SDRi scores on the test set of Libri2Mix. Table 4 reports, for each backbone, the number of utterances falling into various SDRi intervals, and the average SDRi score over the bottom 30% mixtures with the lowest SDRi (denoted as “Tail Performance”), which serves as a proxy for worst-case performance. Every model is evaluated on 6,000 test mixtures.

For Conv-TasNet, the standard dynamic mixing baseline produces 203 mixtures with SDRi below 10 dB, whereas the proposed hard-sampling variant reduces the number to 156. The average SDRi of the bottom 30% mixtures improves from 12.9 to 13.2 dB. Meanwhile, the counts in the mid-to-high intervals (i.e., $[10, 15]$ and $[15, 20]$ dB) slightly increase, while the number of very high SDRi cases ($[20, \infty)$ dB) decreases marginally. This behavior is consistent with the mean SI-SDRi results in Table 1: the curriculum improves the worst part of the distribution without changing the overall average.

For BSRNN, the performance trend is similar but the effect is more favorable. The number of mixtures below 10 dB decreases from 25 to 15, and the average SDRi of the bottom 30% increases from 17.4 to 17.6 dB. In contrast to Conv-TasNet, the number of mixtures with very high SDRi (i.e., $[20, \infty)$ dB)

Table 4: Number of test samples falling into various SDRi intervals, and average SDRi of bottom 30% samples.

Model	Training Strategy	Number of Samples in Different Intervals of SDRi						Tail Performance
		$(-\infty, 0)$	$[0, 5)$	$[5, 10)$	$[10, 15)$	$[15, 20)$	$[20, \infty)$	Bottom30%
Conv-TasNet	Standard	8	59	136	1236	3771	790	12.9
	+ Hard-pair sampling	10	38	108	1311	3823	710	13.2
BSRNN	Standard	2	2	21	132	1896	3947	17.4
	+ Hard-pair sampling	2	1	12	134	1891	3960	17.6
TF-GridNet	Standard	4	7	17	75	1306	4591	18.3
	+ Hard-pair sampling	2	4	15	83	1167	4729	18.8

also increases slightly (from 3,947 to 3,960), indicating that hard-pair sampling has potential to shift the distribution toward better separation quality without sacrificing the top end.

For TF-GridNet, we observe clean gains. With standard training, there are 27 mixtures with SDRi below 10 dB, while, with hard-pair sampling, this number drops to only 7. At the same time, the number of utterances in the highest interval (i.e., $[20, \infty)$ dB) increases from 4,591 to 4,729. The average SDRi of the bottom 30% mixtures improves notably from 18.3 to 18.8 dB, showing that the proposed curriculum substantially mitigates long-tail failures for this backbone.

Overall, these results confirm that curriculum hard-pair sampling consistently improves worst-case performance across all the three DNN models, mainly by reducing the number of poorly separated mixtures and raising the SDRi of the bottom 30%. For stronger backbones such as BSRNN and TF-GridNet, the gains on hard mixtures come without degrading the performance on easier mixtures, and in some cases even slightly improving the performance on easier mixtures, leading to an overall more favorable SDRi distribution.

4.3. Visualization of Speaker-Speaker Difficulty Matrix

To better understand how the proposed sampler interacts with the data, we visualize the speaker-speaker difficulty matrix $\mathbf{D}$ at different training stages. For each matrix, we apply hierarchical clustering [41] to group speakers with similar difficulty profiles. This re-ordering places speakers with similar interaction patterns next to each other, making block structures in Fig. 1 easier to analyze. Entry (i, j) encodes the difficulty score for the speaker pair (s_i, s_j) , with warmer color indicating higher difficulty. The diagonal is masked out because a speaker is never paired with itself.

Panels (a) and (d) correspond to the early stage of training. With standard dynamic mixing and uniform pair sampling (in Fig. 1(a)), we observe several broad, light-colored blocks along the diagonal. These blocks indicate large groups of speakers that are mutually hard to separate from each other, i.e., many pairs inside each block receive high difficulty scores. When curriculum hard-pair sampling is enabled (in Fig. 1(d)), the pattern becomes more concentrated: high-difficulty regions are confined to a few sharper clusters, while most pixels in the matrix is dominated by cooler colors. This suggests that the sampler quickly finds a small set of difficult speakers and prioritizes mixtures involving them in early training, rather than distributing hard mixtures evenly across all speakers.

Panels (b) and (e) show the matrices in the middle of the training process, after the models have largely converged on easy and moderately-hard mixtures. For dynamic mixing with uniform sampling (Fig. 1(b)), the matrix appears more homogeneous than that in the early stage, but moderately difficult regions still spread across many speakers. In contrast, with curriculum

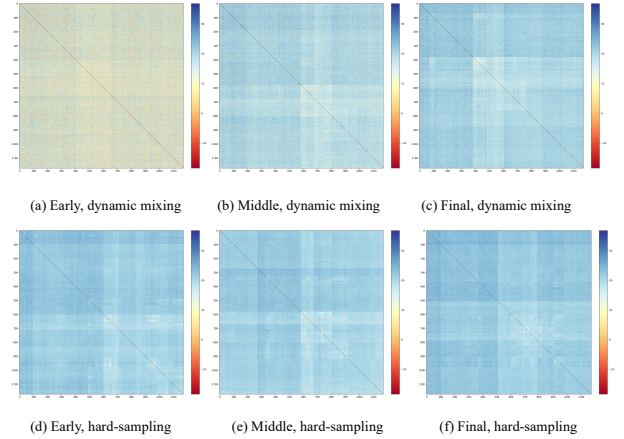


Figure 1: Illustration of speaker-speaker difficulty matrices at different training stages.

hard-pair sampling (in Fig. 1(e)), the high-difficulty mass collapses into a small set of rows and columns. A cross-shaped pattern emerges around a compact speaker cluster: only pairs involving this cluster remain consistently hard, while the rest of the matrix is dominated by lower difficulty scores. This indicates that, by the middle of the training process, the sampler has effectively eliminated most hard pairs and isolated a small, core set of truly challenging speakers that continue to receive emphasis during training.

Overall, the visualization supports the quantitative results reported in Section 4.1 and 4.2. We highlight that curriculum hard-pair sampling does not create many new hard mixtures, but, instead, concentrates the residual difficulty into a small portion of the speaker-speaker space, while reducing difficulty for the vast majority of pairs. This explains why the method improves tail performance and reduces the number of severely under-separated mixtures without sacrificing the mean SI-SDRi.

5. Conclusions

We have proposed a curriculum hard-pair sampling method for speech separation. It builds an online speaker-speaker difficulty matrix during training and gradually shifts the sampling process from near-uniform sampling to sampling harder speaker pairs more. In this way, the model can first learn from common mixtures and then focus more on difficult cases. Evaluation results on Libri2Mix show that the proposed method consistently improves tail performance for Conv-TasNet, BSRNN, and TF-GridNet. In particular, it reduced the number of poorly separated mixtures and improved the SDRi of the bottom 30% samples, while preserving competitive average performance. These results indicate that it is a simple and effective way to improve the robustness of speech separation systems.

6. Generative AI Use Disclosure

Generative AI was used only for language polishing. The authors produced all content and results, and take full responsibility for the paper.

7. References

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